VIBRATION AND STRUCTURAL DYNAMICS

PROFESSIONAL ELECTIVE – I

VI Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6AE27	PEC	3	0	0	3	40	60	100

UNIT-I INTRODUCTION AND SINGLE DEGREE OF FREEDOM SYSTEMS

INTRODUCTION: Simple harmonic motion, terminology, Newton's Law, D'Alembert's Principle, Resonance, Introduction to mechanism of damping. Oscillations. Degrees of freedom.Various mechanisms of damping. Equivalent stiffness concept, vibration isolation

SINGLE DEGREE OF FREEDOM SYSTEMS: Free vibrations – damped vibrations– forced vibrations, with and without damping – transmissibility – vibration measuring instruments.

UNIT-II TWO DOF &MULTI DOFSYSTEMS AND VIBRATION OF CONTINUOUS SYSTEMS

Two degrees of freedom systems (linear and rotating systems), Semidefinite systems coordinate coupling - static and dynamic couplings, vibration absorbers and their applications, Multi degree of freedom systems - principal co-ordinates - principal modes.

Vibrations of continuous systems- Transverse vibration of beams, Longitudinal vibration of bars, lateral vibration of a string, torsional vibration of uniform shafts.

UNIT-III NATURAL FREQUENCIES AND ROTATING SHAFTS

Determining natural frequencies and mode shape (linear and rotating systems), Determination of Eigen Values and Eigen modes – Matrix method.

Natural frequency of rotating shafts Whirling of shafts. Dynamic balancing of rotating shafts. Dynamic dampers.

UNIT-IV APPROXIMATE METHODS FOR FREQUENCY

Introduction to approximate methods for frequency analysis- Dunkerley, Rayleigh's method, Holzer's method, Studola's method Matrix iteration methods. Rayleigh Ritz method for vibration analysis. Diagonalization of stiffness, mass and damping matrices using orthogonality conditions.

UNIT-V THEORY OF AEROELASTIC STABILITY

Aeroelastic and inertial coupling- aeroelastic problems. Collar's triangle. Static and dynamic aeroelastic phenomena. Aeroelastic instabilities and their prevention. Wing divergence, control reversal and wing flutter–buffeting, flutter speed. Aeroelastic tailoring. Elements of servo elasticity

Text Books:

1.Mechanical Vibrations by V. Singh
2. Fug Y. C. (2008), An Introduction to Theory of Aeroelasticity, Dover Publications, US

Reference Books:

1. J. S. Rao, Gupta K. (2002), Theory and practice of Mechanical vibrations, Wiley Eastern Ltd, USA
2. Megson T. H. G (2012), Aircraft Structures for Engineering Students, 5th edition, Elsevier, New York

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Analyze the vibration problems and obtain the natural frequency for the linear and rotatory single-degree-of-freedom systems.
2. "Analyze the vibrating system and estimate the governing equation of motion for twoand multi-degree-of-freedom systems vibration of continuous systems"
3. "Determine the natural frequencies and mode shapes for the spring-mass system androtating shafts"
4. "Evaluate the approximate methods for frequency analysis and obtain the naturalfrequency using therayleigh ritz method for vibration analysis."
5. Analyze the static and dynamic aeroelastic problems associated with aeroelasticity.